Scalable, Secure, Cost-Effective: An End-to-End Edge-Cloud Distributed Solution for Enterprise AI Inference

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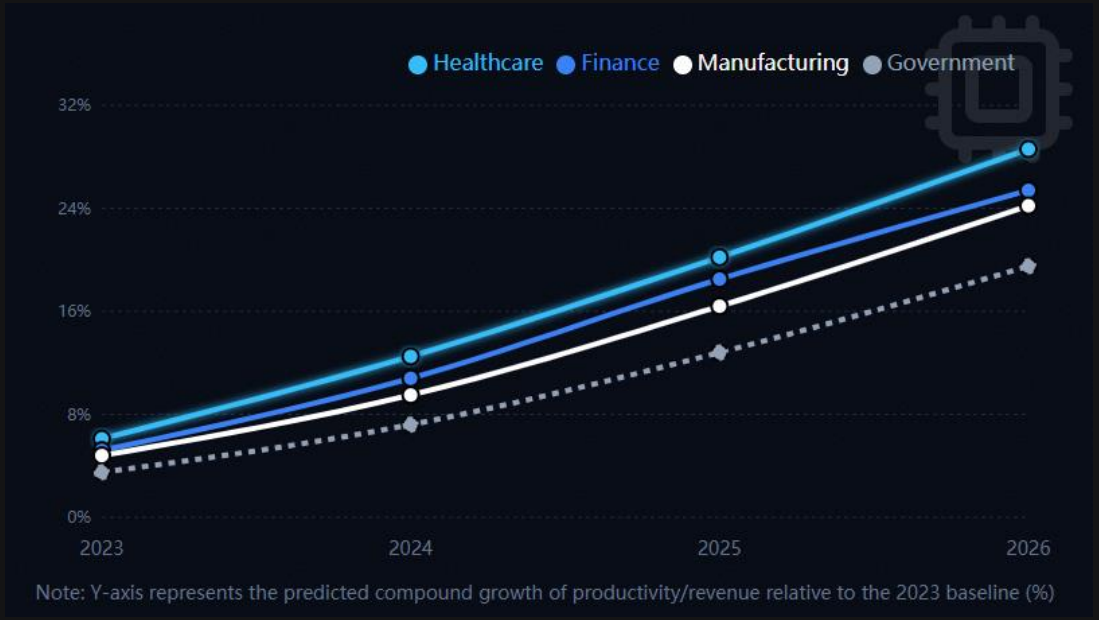
GenAI Industry Growth & Private Deployment Trends



Finance
PRIVATE DEPLOYMENT INDEX **Critical**
"Regulatory compliance, AML risk control, and extreme data privacy requirements."

Healthcare
PRIVATE DEPLOYMENT INDEX **Critical**
"Patient privacy protection (HIPAA, etc.) and sensitive clinical research data."

Government
PRIVATE DEPLOYMENT INDEX **Critical**
"Data sovereignty, national security, and public service compliance."

Manufacturing
PRIVATE DEPLOYMENT INDEX **High**
"Protection of Industrial IP, patent processes, and low-latency edge computing needs."

~28%
MAX GROWTH (2026)

85%+
PRIVATIZATION RATE

AI Evolution & Productivity Surge

AI technology is advancing exponentially, with computing power and model capabilities (LLMs, multimodal) reaching new heights. 2026 is the turning point for GenAI moving from "experimentation" to "production environments."

Enterprise Demand & Workflow Transformation

AI Agents are reshaping workflows for higher automation. Due to strict data privacy and compliance needs, demand for private AI inference deployments is surging, led by healthcare, finance, government, and manufacturing.

The Burden of On-Premises AI Infrastructure

High Initial Investment

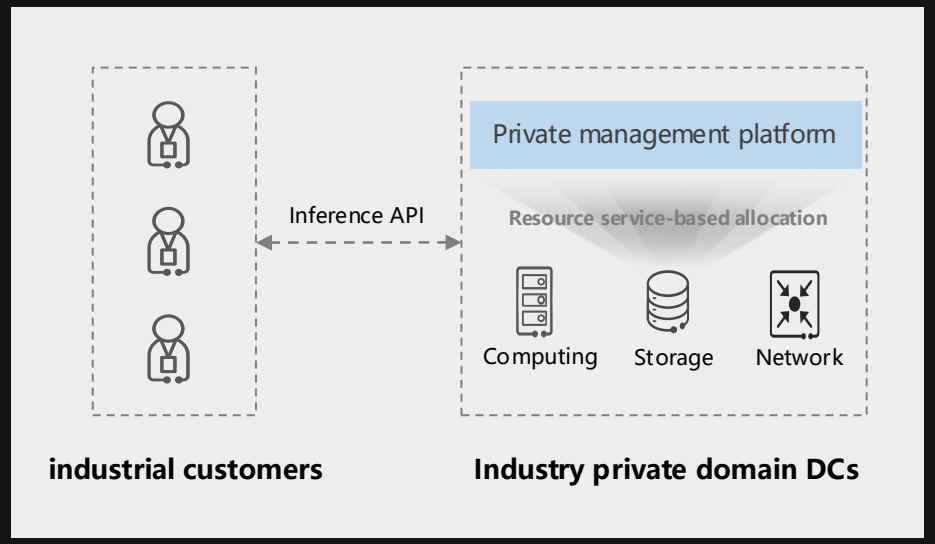
Building an on-premises AI infrastructure requires a massive upfront capital expenditure on servers, GPUs, and other hardware. **(>$500K)**

Prohibitive Operational Costs

Ongoing maintenance, software updates, and the need for specialized IT staff create a heavy financial burden. **(consume 30% IT budget)**

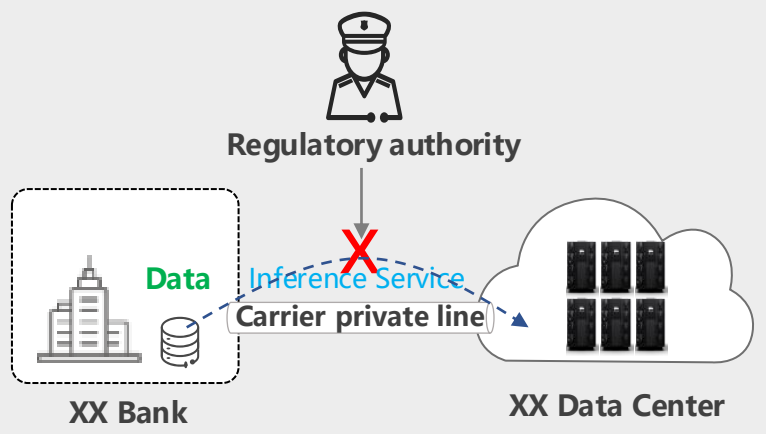
Severe Resource Underutilization

AI inference workloads are often variable, leading to significant hardware idle time during off-peak periods, resulting in poor ROI. **(avg.< 65%)**



Enterprises face a structural paradox in private cluster deployment: incomplete infrastructure, high maintenance costs, and difficult capacity expansion.

Data Privacy & Compliance Risks in Public Cloud Leasing



Example: A bank project was stopped by the regulatory authority due to data security requirements.

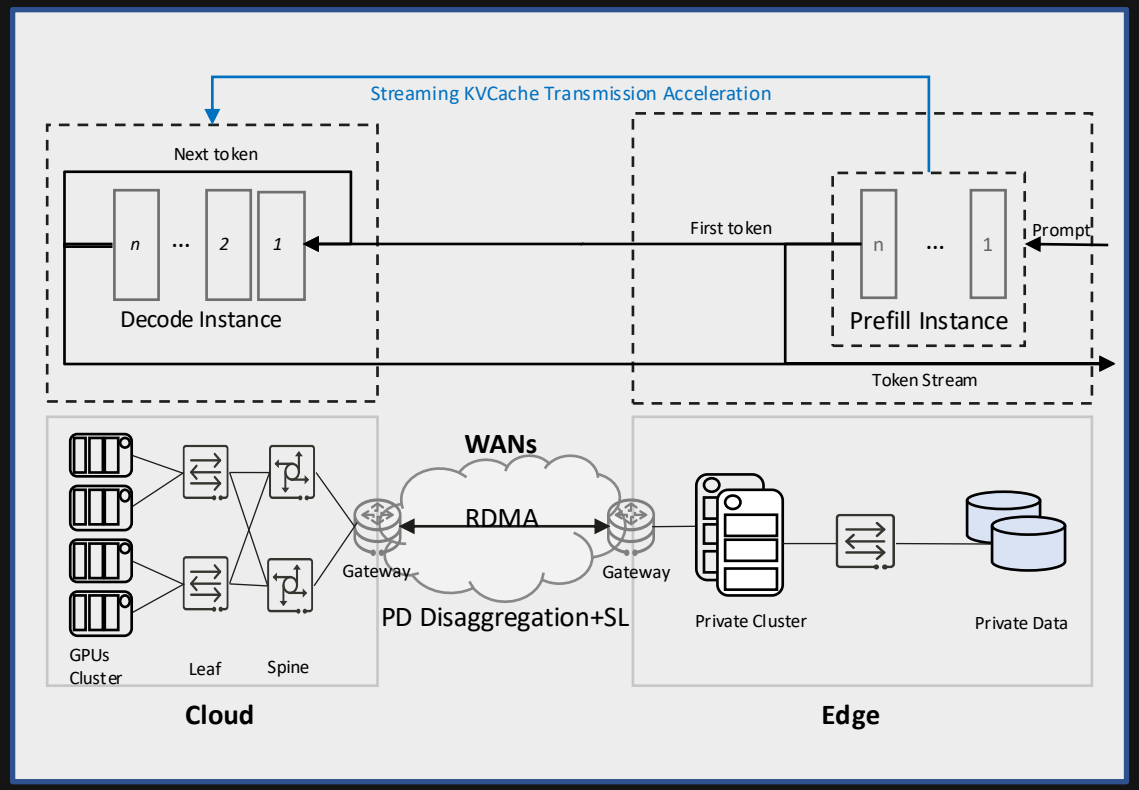
Critical Data Privacy Concerns

Leasing computing power from public clouds often requires uploading sensitive business data, exposing it to risks of breaches and unauthorized access.

Stringent Regulatory Compliance Challenges

Many industries and regions have strict data residency and privacy laws (like GDPR), which public cloud services may struggle to fully comply with, putting businesses at legal risk.

Innovative Distributed Inference Architecture (DIN)



P/D Disaggregation

Separates prefill and decode for optimizing resource utilization and flexibility. Seamless session-state migration across nodes via KV Cache Roaming.

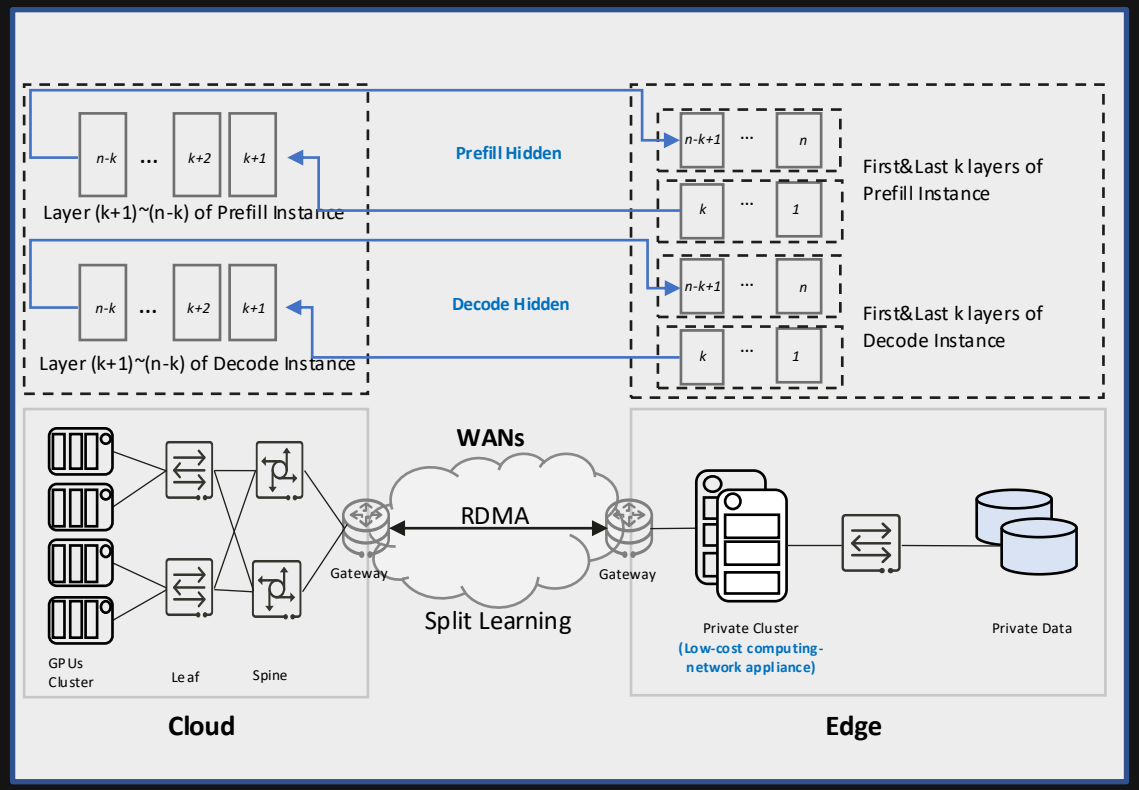
Elastic Cloud-Side Scaling

Dynamically scales cloud-side decoding capabilities based on real-time demand, ensuring efficient resource allocation.

Lossless Network Transmission

Employs RDMA protocols to ensure high-speed, reliable, and lossless data (KVCache) transfer & inference accuracy between edge and cloud.

Edge-Cloud Collaborative Deployment & Security Assurance



Edge Security

Layered deployment localizes first/last layer computation. Only high-dimensional vectors exit the edge, preventing data reconstruction.

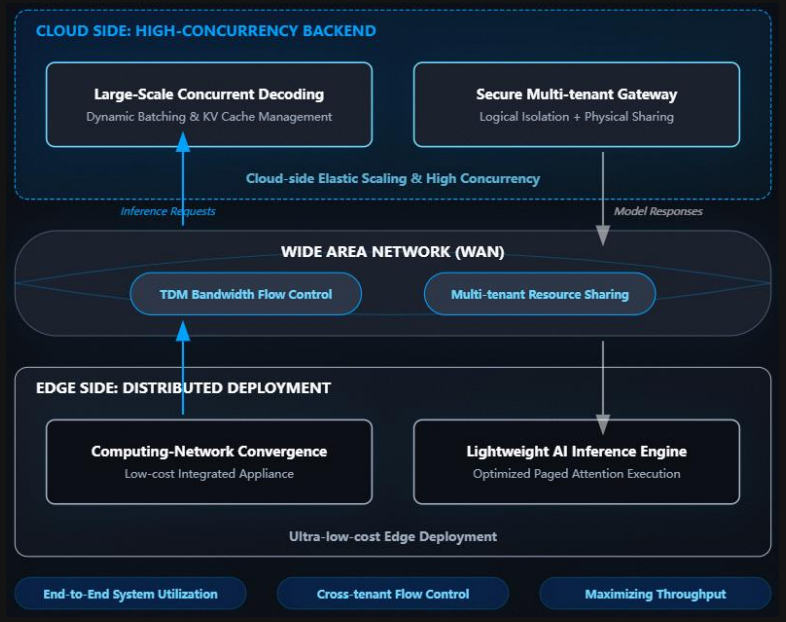
Network Efficiency

Computing-WAN integration optimization tech achieves **95% compute efficiency** and a **320:1 convergence ratio**, maximizing throughput.

Intelligent CPE

"One-Click" deployment integrating RoCE and IPv6+ iFIT for simplified maintenance and lossless AI traffic.

Future Trends: Next-Generation AI Inference



Trend 1: Ultra-Low Cost Edge Computing

- The rise of computing-network integrated hardware will further reduce the cost of edge deployments.
- Shared bandwidth and cloud-side resource pooling will maximize the utilization of computing power, driving down costs.

Trend 2: Long-Term Memory & Compute-Storage Separ

- Future systems will manage private data at the edge with long-term memory capabilities, enabling models to learn over time.
- Separating compute and storage with on-demand preloading will efficiently support larger, more complex AI models.

Conclusion: The Future is Distributed

Inevitable Trend

Private AI model deployment is essential for enterprises to leverage AI while maintaining data control. **Distributed inference**, combining edge-cloud collaboration, offers the optimal balance of cost, flexibility, and security.

Future Direction

The industry is moving towards **'compute via transmission'** and extreme resource efficiency. Driven by long-term memory needs and breaking cloud-side bottlenecks, secure sharing of compute, network, and storage resources across tenants is the next frontier.

Key Takeaway: Adopting edge-cloud distributed AI is not just about solving today's challenges, but securing a competitive advantage in the AI-driven future.

Thank you.

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